Daniel Rhodes

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Profile

MSc Transport Planning student at University of Westminster with a BEng in Electrical and Electronic Engineering from Imperial College London. Experienced in research of London's public transport network, including modelling the network in Python. Seeking a graduate urban planning or transport planning role starting September 2027.

Education

University of Westminster *September 2026 – August 2027*
MSc Transport Planning

Imperial College London *September 2023 – August 2026*
BEng Electrical and Electronic Engineering, 2:1

Gosforth Academy *September 2021 – August 2023*
A levels: A*, A*, A*, A in Maths, Further Maths, Physics and Music respectively

Professional Experience

Undergraduate Research Project, Imperial College London *July – September 2025*

- Applied graph theory using Python (pandas, NetworkX) to London's public transport network, modelled as a graph of around 500 nodes and 700 edges
- Assessed connectivity and resilience using metrics such as betweenness centrality and average path length
- Analysed 8 scenarios to evaluate how changes in network structure affect resilience to failures
- Found that proposed new lines and the Superloop significantly improved network connectivity and robustness
- Planned and delivered the project over 8 weeks, with findings presented in an academic poster

Product Design and Production, Northern CNC *September 2023*

- Designed and produced logos, signs and custom components, turning design concepts into finished products
- Operated laser cutters and mechanical saws safely, carrying out quality checks on finished items

In-Store Team Member, Domino's *August 2021 – December 2022*

- Took and prepared customer orders in a fast-paced environment, working with colleagues during busy periods and following food hygiene and safety procedures

Other Experience

Tube Station Photography, Independent Project *Ongoing*

- Visiting every London Underground station to photograph the station and the surrounding architecture. Full project on the website at the top of the page
- Cycling around London to observe infrastructure first-hand such as cycleways, junction layouts and pedestrian infrastructure

Chess Endgame Analysis Engine, Imperial College London *October 2025 – July 2026*

- Built a chess engine in C++ to detect and evaluate fortress positions, a type of defensive draw that mainstream engines such as Stockfish often misjudge
- Combined exhaustive search, rising-evaluation detection and Stockfish-adapted heuristics; produced a catalogue of fortress positions, an academic poster and a written report

Real Time Grip Strength Tester, Imperial College London *February – March 2026*

- Developed code and a physical device for testing grip strength in real time, using C++ and an Arduino

Second Year Group Project, Imperial College London *June – July 2025*

- Produced a small-scale smart grid that reacted to real time demand and allocated energy usage accordingly

Fundraising, National Citizenship Service *August 2022*

- Raised over £1000 for Mind during a group project with the National Citizenship Service

Piano Accompanist, Local Choir *September 2019 – March 2020*

- Learned many piano pieces in a short time to accompany a local choir, resulting in a successful concert

Skills and Achievements

- **Programming:** Python (pandas, NetworkX), C++, MATLAB, HTML
- **Software:** Excel, SPSS, ArcGIS, Qualtrics
- **Languages:** French (B1), German (A2)
- **ABRSM:** ARSM diploma in piano (distinction), distinction in grade 8 piano, clarinet and music theory